

LPWA Tutored Work Correction

Requirements:

- LoRa Modem Calculator
- LoRa Alliance Bands Allocation

Exercise1:

Considering the spreading factors listed in the table, calculate the user data bits and the CRC dedicated ones according to the four coding rates.

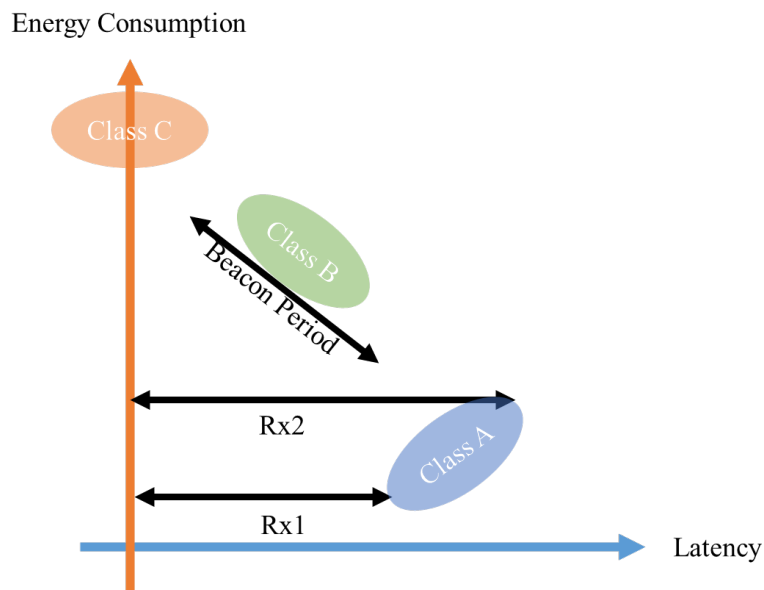
$$\text{User Data} = \text{SF} * (4 / (4 + \text{CR}))$$

$$\text{CRC} = \text{SF} - \text{User Data}$$

SF	Bits	Coding Rate			
		1	2	3	4
7	User Data	5.6	4.6	4	3.5
	CRC	1.4	2.4	3	3.5
9	User Data	7.2	6	5.14	4.5
	CRC	1.8	3	3.86	4.5
11	User Data	8.8	7.3	6.29	5.5
	CRC	2.2	3.7	4.71	5.5

Exercise2 :

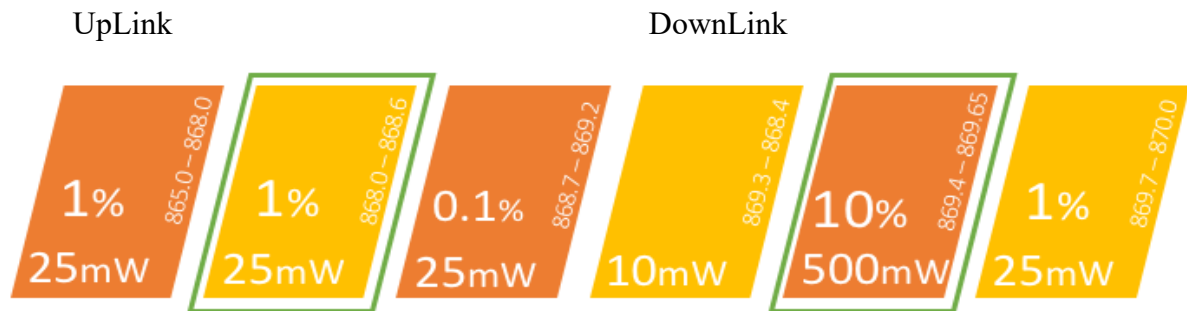
Make a graph plotting the energy consumption according to the reception latency of the LoRa compatible device classes.



Exercise3 :

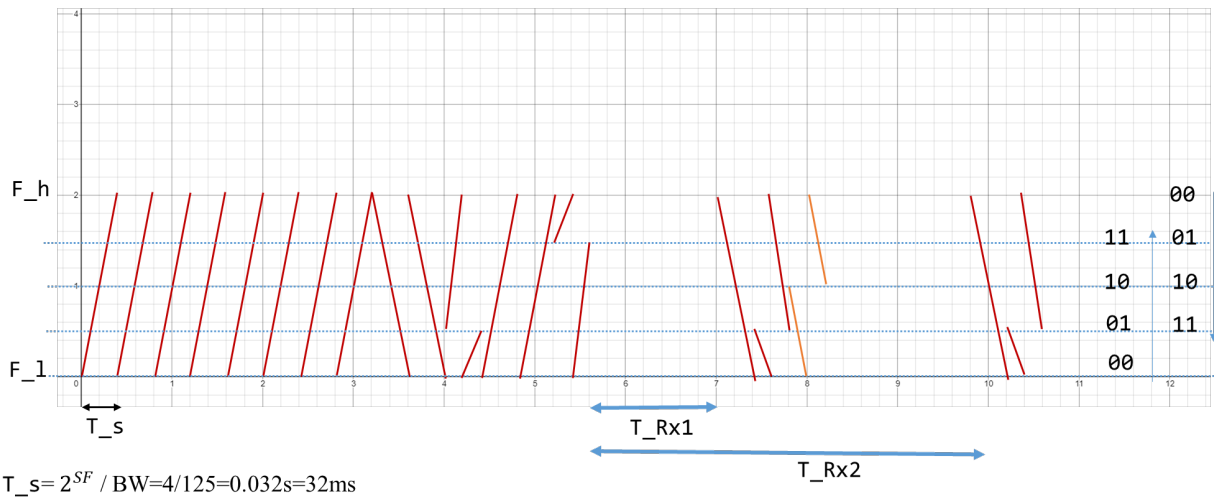
Considering a Class A device sends a payload of 8 bits “01000011” through the LoRaWAN technology. The device receives a payload of 4 bits “0011” from the gateway to acknowledge the reception. You are asked to draw the shape of the modulated RF signal to make the discussed communication. To simplify, you can use a spreading factor of 2 and 8 preambles with a bandwidth of 125 KHz.

You can choose one of the following LoRa bands:



“01.00.00.11”

“00.11.10”
‘10’ (orange) term is extra in this case to show how to make it



Pay attention to the downlink order, in this case is changed : we start 00 at Frequency high and we end with 11 at level before Frequency Low.

In the exam, you will get used convention for UP/DOWN links.

Exercise4: This exercise needs the LoRA Calculator, so it will not be included in the exam. Good Luck for you all.

The objectif of this exercise is to study the EU433 and EU868 bands effect on the battery lifetime. We consider a device with lithium battery of 5000mAh and 3.7 operating voltage. The preamble for EU433 is 5 symbols and 8 for EU868. The bandwidth is 125 KHz and the SF is 8. The payload is 10 Bytes with a coding rate is 1. The header mode is set to default and the device is sending 4 msgs per day. The duty cycle for both bands is 1%, but the power of EU433 is 13dBm and 20dBm for EU868 one.

Make the simulations to calculate the timing performance indicators then the energy-based ones

You can remark that the EU433 makes the double of the battery lifetime compared to EU868